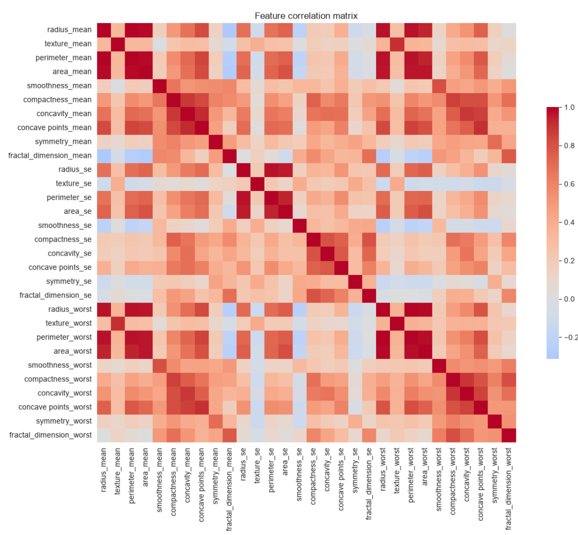
Breast Cancer Detection with a PCA/NMF Ensemble Classifier

Fusing complementary feature representations and classifiers to flag malignant tumours reliably.

Jibril Hussaini | Python · scikit-learn · NumPy · pandas

Overview

Early, accurate detection of malignant tumours is one of the highest-value applications of machine learning in healthcare. This project builds a diagnostic classifier for the Breast Cancer Wisconsin (Diagnostic) dataset — 569 tumours, each described by 30 numeric measurements and labelled benign or malignant. Rather than relying on a single model, it combines two feature-extraction methods (Principal Component Analysis and Non-negative Matrix Factorization) with two classifiers (Logistic Regression and a Support Vector Machine), and merges the four resulting predictors by majority voting.



Test-set confusion matrix for the selected ensemble.

Goals

Deliver a robust, well-validated detector that maximises recall on malignant cases without sacrificing precision, and demonstrate that an ensemble of complementary feature/classifier pairings outperforms any single member.

Objectives

- Build an end-to-end pipeline: stratified 60/20/20 train/validation/test split, Min-Max feature scaling fitted on the training set only, and dimensionality reduction with PCA and NMF.
- Train four base learners — (PCA, LR), (NMF, LR), (PCA, SVM), (NMF, SVM) — and combine them by majority vote, resolving ties toward the malignant class to minimise dangerous false negatives.

- Tune the number of extracted components (2, 4, 6, 8, and 10) on the validation set and select the best configuration before touching the test set.
- Evaluate the final model on a held-out test set using F-score, accuracy, precision, and recall.

Approach

Exploratory analysis showed the 30 features are highly redundant (size measurements correlate above 0.99), so the data effectively lives in far fewer dimensions — an ideal setting for PCA and NMF. For each candidate k, the four base learners are trained with 5-fold cross-validation and their validation predictions are combined by majority vote. The configuration with the highest validation F-score is locked in and evaluated once on the test set for an unbiased performance estimate.

Outcomes

- The ten-component ensemble was selected and reached an F-score of 0.976 and accuracy of 0.982 on the held-out test set.
- Precision of 1.000 (no false positives) with recall of 0.952 — only two malignant cases missed out of 42.
- Majority voting absorbed the weakest member (NMF + Logistic Regression) without losing accuracy, confirming the robustness benefit of the ensemble.
- Identified threshold tuning and class-sensitive classifiers as the clearest path to closing the remaining two false negatives.